

DBMS COURSE PROJECT · TEAM 1

MediCare

Hospital Information and Management Database

Eleven connected tables bringing patient records, doctors, appointments, admissions, prescriptions and billing into a single relational system — replacing duplicate paper records and double-booked rooms.

11

Tables

11

Business queries

6

Team members

MySQL 8.0+

RDBMS

Problem Understanding & Requirements

Hospital records scattered across paper files and spreadsheets cause duplicate patients, double-booked rooms and billing that doesn't match the care given. MediCare centralises all of it into one connected system.

Patient Management

Unique patient_id, lookup before re-registering, full visit history.

Doctors & Departments

Each doctor in exactly one department, with specialisation and fee.

Appointments

Scheduled, Completed or Cancelled; diagnoses tied to each visit.

Admissions & Rooms

Room inventory and rates; one active admission per room.

Pharmacy

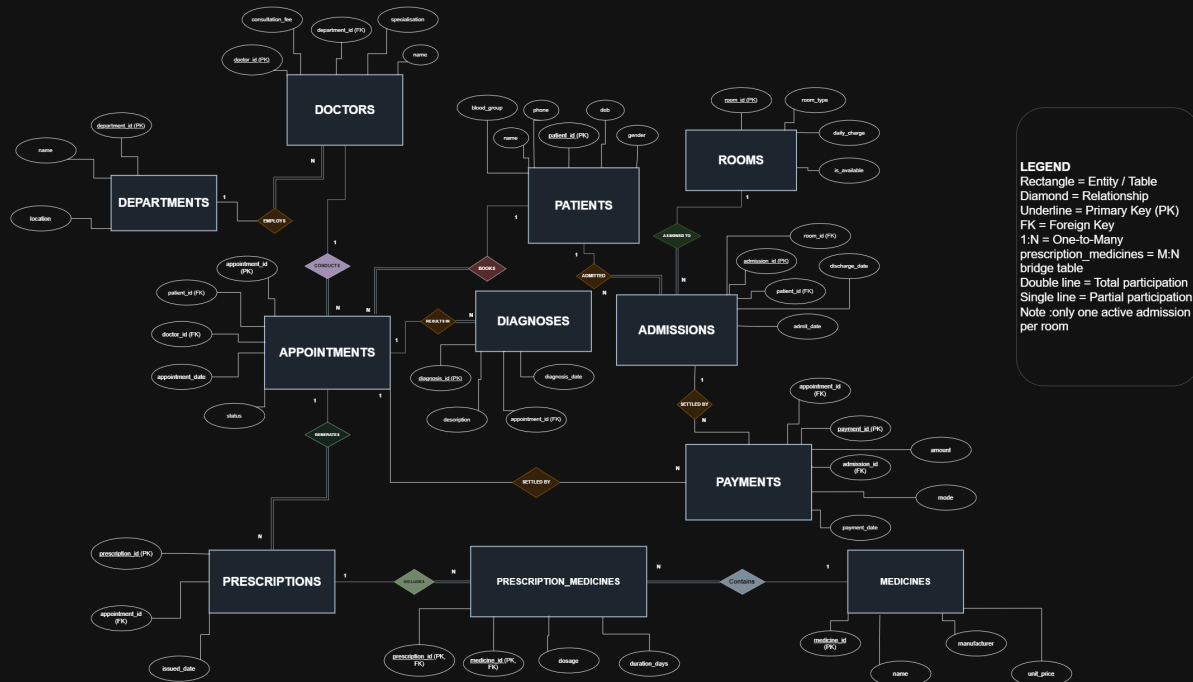
A prescription can list many medicines, from a central catalog.

Billing

Every payment traces to one appointment or one admission.

ER Diagram & Relational Design

MediCare — Entity Relationship Diagram



11 entities, Chen notation

Rectangles are tables, diamonds are relationships, underlines mark primary keys.

1:N everywhere except one

Department → Doctors, Patient → Appointments/ Admissions, Appointment → Diagnoses/Prescriptions.

M:N resolved with a bridge

prescription_medicines carries the composite key for prescriptions ↔ medicines.

One rule worth noting

A payment settles exactly one appointment or one admission — never both.

Normalisation & Database Structure

All 11 relations verified up to Third Normal Form, starting from one unnormalised hospital record.

1NF

Atomic values, no repeating groups — multi-medicine prescriptions split one row per prescription-medicine pair.

2NF

No partial dependency. Only `prescription_medicines` has a composite key (`prescription_id`, `medicine_id`) — dosage and `duration_days` depend on the whole key.

3NF

No transitive dependency. Department name lives only in departments, not copied onto every doctor row — same for rooms, medicines and payments.

ANOMALIES AVOIDED

Update — master data edited once, not across every transaction row · Insert — a doctor or medicine exists independently · Delete — removing an appointment never removes doctor, patient or department data

Database Implementation & Data Quality

SAMPLE DATA — VERIFIED ROW COUNTS

5 Depts	9 Doctors	20 Patients	8 Rooms
32 Appts	10 Admissions	29 Diagnoses	12 Medicines
29 Prescriptions	39 Rx-Medicines	37 Payments	0 load errors

Constraints, not just columns

NOT NULL, UNIQUE, CHECK (status, gender, positive amounts, discharge $\geq$ admit), DEFAULT, and FK actions chosen per relationship — RESTRICT on master data, CASCADE on dependent detail rows.

Data built on purpose, not at random

Patients with zero visits, repeat admissions, skewed medicine frequency, a 6-month payment spread — every business query has a real answer to find.

6 data-quality issues found & fixed

Misrouted prescriptions, a double-booked room, wrong room-status flags, two incorrect bills, mismatched doctors, and pediatric cases on adult patients — each traced, corrected, and re-verified.

SQL Queries & Analysis

REQUIRED — Q1 TO Q7

- Q1** Doctors who treated the most patients — COUNT(DISTINCT)
- Q2** Busiest department by appointment count
- Q3** Average consultation fee by department
- Q4** Patients who never visited — LEFT JOIN ... IS NULL
- Q5** Most-prescribed medicines — via the bridge table
- Q6** Patients admitted more than once — HAVING COUNT > 1
- Q7** Monthly revenue trend — DATE_FORMAT + SUM

TEAM'S OWN — Q8 TO Q11

- Q8** Room occupancy — times each room was used
- Q9** Average length of stay per room type — DATEDIFF
- Q10** Estimated consultation revenue per doctor
- Q11** Cancellation rate per department — CASE WHEN

All 11 verified live

Every query run against MySQL 8.0 — non-empty, non-tied, correct results.

GitHub Repository & Documentation

MediCare-main/

└ README.md

└ MediCare_Project_Report.md + .pdf

└ database/

 create_tables · insert_data · data_fixes

└ data queries/queries.sql

└ diagrams/

 ER diagram · relational schema

└ docs/

 requirements · normalisation · design rationale

One README, orients in 30 seconds

Project description, full team list with roles, setup commands, folder map.

Every section stitched, not rewritten

Requirements, ER design, normalisation, implementation and testing kept in each author's own voice.

Staged commit history

Structure → README → diagrams → schema → data → queries
→ docs, so the repo reads as real progress.

Project Report & Code Testing

TEST PASSES — CLEAN EVERY TIME

- Schema script runs on a clean MySQL 8.0 database — 11 tables, every constraint, zero errors
- Data script loads all 11 tables in dependency order, zero errors
- All 11 queries run clean, non-empty, non-tied results
- Each of the 6 data fixes re-verified with its own SQL check after correction

Project report

Full write-up covering every stage — problem, design, normalisation, implementation, queries — delivered as both Markdown and a formatted PDF.

Testing approach

Nothing accepted on a first pass — every script and every fix re-run against a live database before being called done.